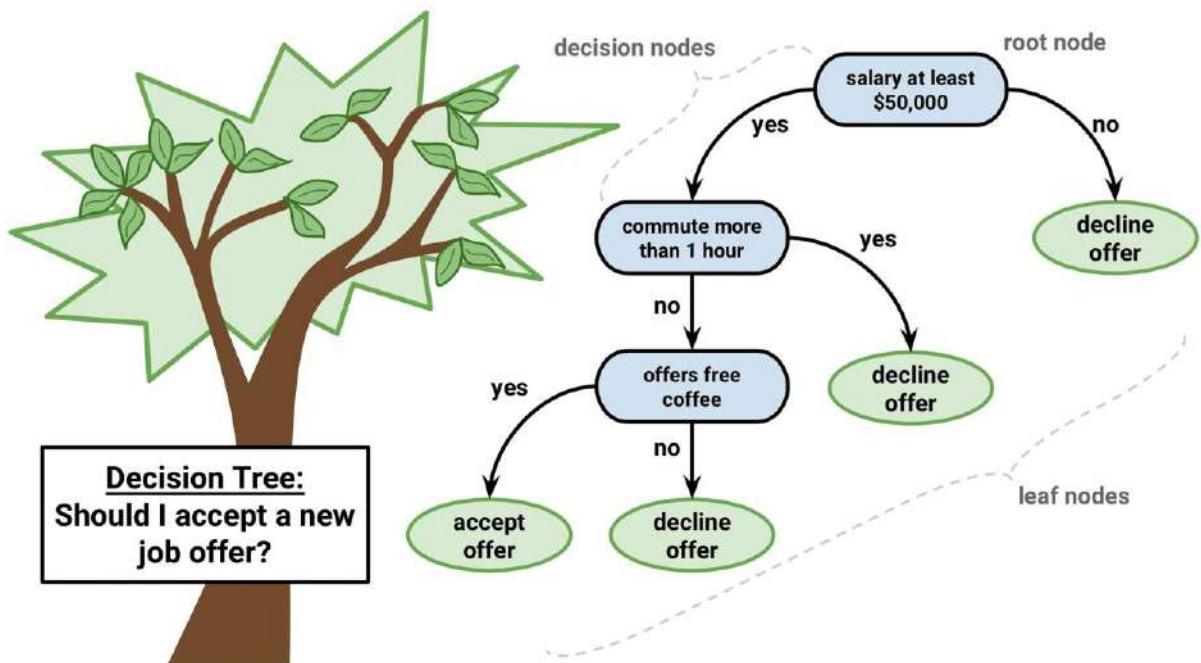


classification Algorithms

Decision Tree:

- It is also a classification algorithms
- *It is like a tree structure*
- *Tree has branches, leafs, roots*
- *Root is the very very important for tree*
- *From roots only we can make branches*
- *Finally from branches we can get leafs*
- *In the Decision Tree Roots and Branches means input features*
- *Leafs means output data lables*



age gender income Cr Buy – a – computer

In Decision Tree The main Goal is : Idenitfy the Root node

which column I need to choose to start with Decision, That is Called Root Node features

Root Node : Good Leader

Good leader : Focus is More

BAD leader : His Mindset is not clear (More random guy)

Root node behaviour: Random ness should be very very less

Ajay and Aravind very very close friends

There is a Family function going on at ajay home

For that ajay and aravind working together

ajay is calling night time 2am is calling

He knows the information ===== No random ness

When you dont know the information ===== More random ness

ind vs Aus : No information(proper) ===== More random ness

ind vs zim:

This concept is identified by a scientist : Shannon Information criteria

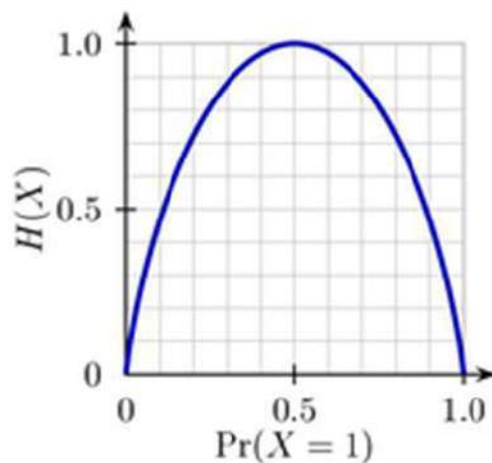
Entropy:

$$H = - \sum_{i=1}^n p_i * \log_2 p_i$$

Entropy gives an Idea ,

- *If we dont know about information, The H will be high*
- *Which indicates more random ness*
- *Where p = probability*
- *$\log_2 p_i$ base: 2 is because information passes in the form of bits*

Information Entropy



- For a Bernoulli trial ($X = \{0, 1\}$) the graph of entropy vs. $\Pr(X = 1)$. The highest $H(X) = 1 = \log(2)$

Toss a Coin

Two possible outcome Head, Tail

$$P(H) = 0.5, \quad P(T) = 0.5$$

$$H = - \sum_{i=1}^n p_i * \log_2 p_i$$

$$H = - \sum_{i=1}^{n=2} p_i * \log_2 p_i = -p_1 * \log_2 p_1 - p_2 * \log_2 p_2$$

$$H = -p_h * \log_2 p_h - p_t * \log_2 p_t$$

$$H = -0.5 * \log_2 0.5 - 0.5 * \log_2 0.5$$

$$H = -0.5 * \log_2 \frac{1}{2} - 0.5 * \log_2 \frac{1}{2} \quad \frac{1}{2} = 2^{-1}$$

$$H = -0.5 * \log_2 2^{-1} - 0.5 * \log_2 2^{-1} \quad \log_a a^x = x * \log_a a$$

$$H = -0.5 * (-1) * \log_2 2 - 0.5 * (-1) * \log_2 2 \quad \log_a a = 1$$

$$H = 0.5 * 1 + 0.5 * 1$$

$$H = 0.5 + 0.5 = 1$$

Hence entropy concept is Proved

More randomness === Not Proper information

More randomness === More entropy

Not proper information === More Randomness === More entropy

In the Decision Tree ,we need a Root node

- *Root node should have proper information*
- *Root node should have less randomness*
- *Root node should have less entropy*
- *we will calculate ,Entropy for each variable*
- *Which variable has less Entropy ,That is our Root Node*

Process

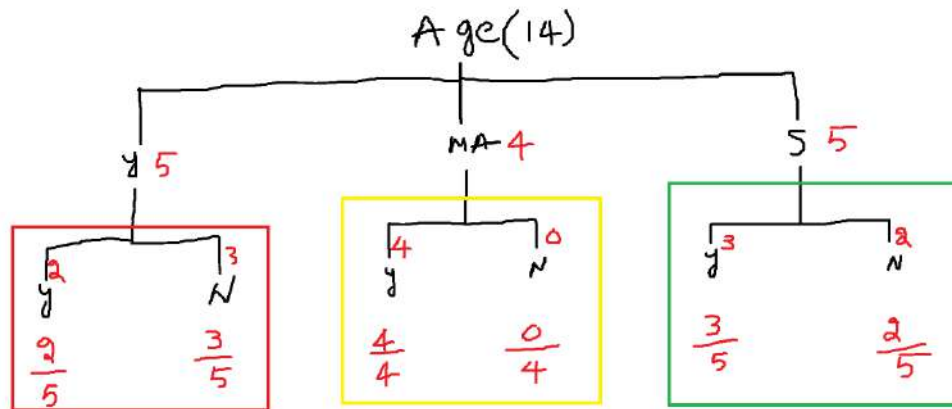
<i>Id</i>	<i>Age</i>	<i>Buys a computer(Target)</i>
1	Youth	No
2	Youth	No
3	Middle Aged	Yes
4	Senior	Yes
5	Senior	Yes
6	Senior	No
7	Middle Aged	Yes
8	Youth	No
9	Youth	Yes
10	Senior	Yes
11	Youth	Yes
12	Middle Aged	Yes
13	Middle Aged	Yes
14	Senior	No

Entropy of Age: $H(\text{Age})$

Age has three labels

- *Youth*
- *Middle Aged*
- *Senior*

$$H(\text{Age}) = H(\text{youth}) + H(\text{MA}) + H(\text{Senior})$$



$$H = - \sum_{i=1}^n p_i * \log_2 p_i$$

H(Youth): Youth have two outcomes, so Two Probabilities

$$H_{youth} = - \sum_{i=1}^{n=2} p_i * \log_2 p_i = -p_1 * \log_2 p_1 - p_2 * \log_2 p_2$$

$$H_{youth} = -p_{yes} * \log_2 p_{yes} - p_{no} * \log_2 p_{no}$$

$$p_{yes} = \frac{2}{5}, \quad p_{no} = \frac{3}{5}$$

$$H_{youth} = -\frac{2}{5} * \log_2 \frac{2}{5} - \frac{3}{5} * \log_2 \frac{3}{5} \quad (\text{Do not calculate answers})$$

H(MA): MA have two outcomes, so Two Probabilities

$$H_{MA} = - \sum_{i=1}^{n=2} p_i * \log_2 p_i = -p_1 * \log_2 p_1 - p_2 * \log_2 p_2$$

$$H_{MA} = -p_{yes} * \log_2 p_{yes} - p_{no} * \log_2 p_{no}$$

$$p_{yes} = \frac{4}{4}, \quad p_{no} = \frac{0}{4}$$

$$H_{MA} = -\frac{4}{4} * \log_2 \frac{4}{4} - \frac{0}{4} * \log_2 \frac{0}{4}$$

H(Senior): Senior have two outcomes, so Two Probabilities

$$H_{Senior} = - \sum_{i=1}^{n=2} p_i * \log_2 p_i = -p_1 * \log_2 p_1 - p_2 * \log_2 p_2$$

$$H_{Senior} = -p_{yes} * \log_2 p_{yes} - p_{no} * \log_2 p_{no}$$

$$p_{yes} = \frac{3}{5}, \quad p_{no} = \frac{2}{5}$$

$$H_{Senior} = -\frac{3}{5} * \log_2 \frac{3}{5} - \frac{2}{5} * \log_2 \frac{2}{5}$$

$$H(Age) = H(youth) + H(MA) + H(Senior)$$

$$\begin{aligned} H(Age) = & \left[-\frac{2}{5} * \log_2 \frac{2}{5} - \frac{3}{5} * \log_2 \frac{3}{5} \right] \\ & + \left[-\frac{4}{4} * \log_2 \frac{4}{4} - \frac{0}{4} * \log_2 \frac{0}{4} \right] \\ & + \left[-\frac{3}{5} * \log_2 \frac{3}{5} - \frac{2}{5} * \log_2 \frac{2}{5} \right] \end{aligned}$$

Weightage concept:

Never add the different group values directly

Provide some weightage

$$H(Age) = W_y * H(youth) + W_{MA} * H(MA) + W_s * H(Senior)$$

$$\begin{aligned} H(Age) = & W_y \left[-\frac{2}{5} * \log_2 \frac{2}{5} - \frac{3}{5} * \log_2 \frac{3}{5} \right] \\ & + W_{MA} \left[-\frac{4}{4} * \log_2 \frac{4}{4} - \frac{0}{4} * \log_2 \frac{0}{4} \right] \\ & + W_s \left[-\frac{3}{5} * \log_2 \frac{3}{5} - \frac{2}{5} * \log_2 \frac{2}{5} \right] \end{aligned}$$

Weightage means Contribution

Out of total data what is the Youth Contribution

There are total 14 observations

$$\text{In that 5 are Youth : } W_y = \frac{5}{14}$$

Out of total data what is the MA Contribution

There are total 14 observations

$$\text{In that 4 are MA : } W_{MA} = \frac{4}{14}$$

Out of total data what is the Senior Contribution

There are total 14 observations

$$\text{In that 5 are Senior : } W_s = \frac{5}{14}$$

$$\begin{aligned} H(\text{Age}) &= \frac{5}{14} * \left[-\frac{2}{5} * \log_2 \frac{2}{5} - \frac{3}{5} * \log_2 \frac{3}{5} \right] \\ &+ \frac{4}{14} * \left[-\frac{4}{4} * \log_2 \frac{4}{4} - \frac{0}{4} * \log_2 \frac{0}{4} \right] \\ &+ \frac{5}{14} * \left[-\frac{3}{5} * \log_2 \frac{3}{5} - \frac{2}{5} * \log_2 \frac{2}{5} \right] \cong 0.7 \end{aligned}$$

Step – 1: Calculate the Individual input feature entropy

- $H(\text{Age}) = W_y * H(\text{youth}) + W_{MA} * H(\text{MA}) + W_s * H(\text{Senior})$
- $H(\text{Income}) = W_H * H(\text{High}) + W_M * H(\text{M}) + W_L * H(\text{Low})$
- $H(\text{student}) = W_{yes} * H(\text{Yes}) + W_{No} * H(\text{No})$
- $H(\text{Credirating}) = W_{Fair} * H(\text{F}) + W_{Excellent} * H(\text{Excellent})$

Assuming

$$H(\text{Age}) = 0.2, H(\text{Income}) = 0.3, H(\text{student}) = 0.4, H(\text{Credirating}) = 0.5$$

Step – 2: We need to calculate total data entropy: Before split

$$\text{Total yes} = 9; \text{Total No} = 5$$

$$p(\text{yes}) = \frac{9}{14}; p(\text{no}) = \frac{5}{14}$$

$$H_{\text{Total}} = - \sum_{i=1}^{n=2} p_i * \log_2 p_i = -p_{\text{yes}} * \log_2 p_{\text{yes}} - p_{\text{no}} * \log_2 p_{\text{no}}$$

$$H_{\text{Total}} = -p_{\text{yes}} * \log_2 p_{\text{yes}} - p_{\text{no}} * \log_2 p_{\text{no}}$$

$$H_{\text{Total}} = -\frac{9}{14} * \log_2 \frac{9}{14} - \frac{5}{14} * \log_2 \frac{5}{14} = 0.940$$

Step – 3: (Information Gain)

- with only entropy we can not identify the root node
- entropy will explain about random ness
- in order to find to root node we need to calculate information gain
- Information gain measures how much uncertainty reduce by a particule feature
- Before split there is so much uncertainty
- Due to feature split ness the uncertainty will be reduces
- Gain will give the idea, which feature reducing more uncertainty
- The feature which has more information gain , that become root node

$$\text{Gain}_{\text{feature}} = H_{\text{Total}} - H_{\text{feature}}$$

$$\text{Gain}_{\text{age}} = H_{\text{Total}} - H_{\text{age}} = 0.9 - 0.7 = 0.2$$

$$\text{Gain}_{\text{income}} = H_{\text{Total}} - H_{\text{income}} = 0.9 - 0.8 = 0.1$$

$$\text{Gain}_{\text{student}} = H_{\text{Total}} - H_{\text{student}} = 0.9 - 0.85 = 0.5$$

$$\text{Gain}_{\text{cr}} = H_{\text{Total}} - H_{\text{cr}} = 0.9 - 0.75 = 0.15$$

Conclusion: In order to get a root Node which variable has highest IG that become root node

1) You should not compare different features on the same level

age vs income vs student vs cr

2) Entropy is tell about random ness : how data is splitted ramdomly

3) Gain is tell about clarity: how a data is splitted well

You can talk in postive way : Gain concept

You can talk in negative way: Entropy

$$\text{Gain}_{\text{feature}} = H_{\text{Total}} - H_{\text{feature}}$$

4) This is we called as Information gain

5) The entire concept name is : **Information gain**

6) The maths we are using in this concept is : **Entropy**

Information Gain:

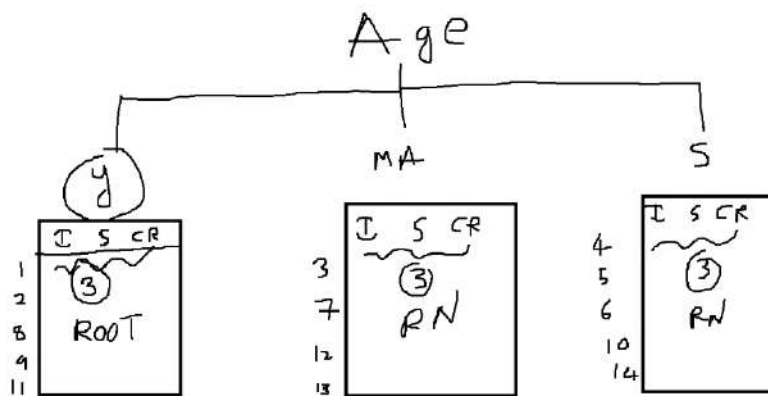
Step – 1: Calculate Individual Feature entropy

Step – 2: Calculate Total data entropy

Step – 3: Calculate the Gain

Conclusion: Which have highest gain, that is the root node

For example assume that : Age is the root node



Every new table also have output column

Draw back:

Information gain fails because of ID type of columns

Calculate entropy for ID column

$$H_{ID} = H_1 + H_2 + \dots + H_{14}$$

$$H(ID) = W_1 * H_1 + W_2 * H_2 \dots W_{14} * H_{14}$$

$$W_1 = W_2 = W_3 = \dots = W_{14} = \frac{1}{14}$$

weighate means contribution

$$H_1 = H_2 = H_3 = \dots = H_{14} = - \sum_{i=1}^{n=1} p_i * \log_2 p_i = - p_1 * \log_2 p_1$$

$$-1 * \log_2 1 = -\log_2 1 = 0$$

$$H(ID) = W_1 * H_1 + W_2 * H_2 \dots W_{14} * H_{14}$$

$$H(ID) = -[\frac{1}{14} * \log_2 1 + \frac{1}{14} * \log_2 1 \dots \frac{1}{14} * \log_2 1]$$

$$H(ID) = \frac{-1}{14} [0 + 0 + \dots + 0] = 0$$

- we already know that for root node , entropy should low and gain will high
- for ID the entropy is zero, Gain = 0.94 - 0 = 0.94
- **Gain is very high**
- if you include ID then automatically , information gain concept consider ID as Root Node
- This leads No pattern learning, Model become overfit
- That's why we always drop the ID column or fetaures which have more lables

Information Gain

Step – 1: Calculate Individual entropy

Step – 2: Calculate total entropy

Step – 3: Calculate gain

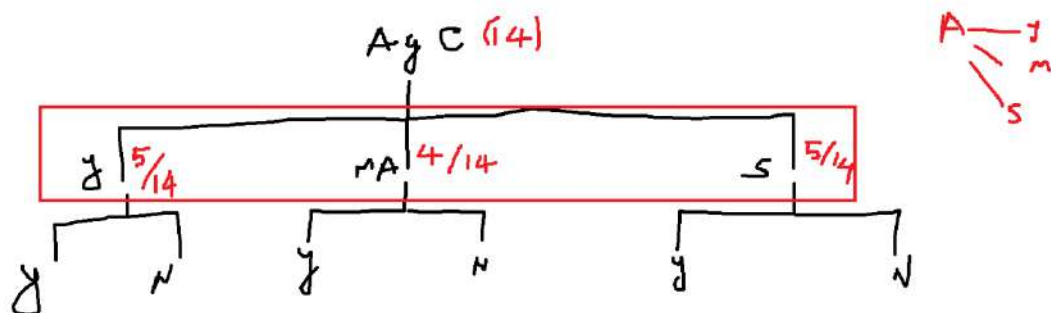
Conclusion: Which feature has heighest gain ,That becomes a root node

Draw back:

- Any feature has more splits ,Gain might be more
- For example ID has more splits, Entropy become zero
- **Gain has more, so we misconclude ID as a Root node**
- Generally we will drop the more splitting columns
- Example id, zip id, phone numbers etc

I dont want drop it, still I want to avoid this draw back

Split Information Ratio:



we need to calculate entropy at split level

Age is divided into 3 splits

- $youth = p(youth) = \frac{5}{14}$
- $MA = p(MA) = \frac{4}{14}$
- $Senior = p(Senior) = \frac{5}{14}$

split information(Age)

$$\begin{aligned}
 &= - \sum_{i=1}^{n=3} p_i * \log_2 p_i = \\
 &= -p_{youth} * \log_2 p_{youth} - p_{ma} * \log_2 p_{ma} - p_s * \log_2 p_s \\
 &= -\frac{5}{14} * \log_2 \frac{5}{14} - \frac{4}{14} * \log_2 \frac{4}{14} - \frac{5}{14} * \log_2 \frac{5}{14} = 0.286
 \end{aligned}$$

Split information will give how broadly the feature is splitted

we want less split information

Step – 1: Calculate Individual entropy

Step – 2: Calculate total entropy

Step – 3: Calculate gain

Step – 4: Split information of each feature

Step – 5: Gain Ratio

$$\text{Gain ratio} = \frac{\text{Gain}}{\text{split information}}$$

$$\text{Gain ratio(Age)} = \frac{\text{Gain(Age)}}{\text{split information(Age)}}$$

$$\text{Gain ratio(income)} = \frac{\text{Gain(income)}}{\text{split information(income)}}$$

$$\text{Gain ratio(Student)} = \frac{\text{Gain(student)}}{\text{split information(student)}}$$

$$\text{Gain ratio(ID)} = \frac{\text{Gain(ID)}}{\text{split information(ID)}}$$

Case – 1:

Gain(Age) = high (assume)

split information(Age) = low (assume)

Gain ratio(Age) = high

$$\text{Gain ratio(Age)} = \frac{\text{Gain(Age)}}{\text{split information(Age)}}$$

Case – 2:

Gain(ID) = high (assume)

split information(Age) = ID (assume)

Gain ratio(ID) = low

$$\text{Gain ratio(ID)} = \frac{\text{Gain(ID)}}{\text{split information(ID)}}$$

- assume that Split information of ID column becomes maximum
- That means denominator is maximum
- then total value less
- **Conclusion: Highest Gain Ratio is our root node**

Information gain

Step – 1: Calculate Individual entropy

Step – 2: Calculate total entropy

Step – 3: Calculate gain = $H_{Total} - H_{feature} = step_2 - step_1$

Step – 4: Split information of each feature

Step – 5: Gain ratio = $\frac{Gain}{Split\ information} = \frac{step3}{step4}$

*entropy: will indicate random ness **this should be low***

*total entropy: will indicate entropy of data(before split) generally **this is high***

gain

*: will indicate reducing the amount of uncertainty while split the data , **this should be high***

*Split information: will indicates how balanced the feature is splitting, **this should be low***

*Gain ratio : it is a normalized form of information gain, **this should be high***

First 3 steps combinley called as ID3 : Iterative dechtomiser

Last 2 steps combinley called as C4.5